Phase 1 · Machine Learning · Project 26

Card Fraud Triage

Finding fraud by learning what normal looks like

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The problem: fraud is rare and hidden

Very rare

Only 17 in every 10,000 transactions are fraud.

Found late

We learn it was fraud days or weeks later.

Always changing

Fraudsters keep changing their tricks.

Accuracy is a trap

Saying “everything is normal” is 99.83% correct. But it finds zero fraud.

Our idea: learn what normal looks like

1. Learn normal

Train only on normal transactions.

2. Score each one

The stranger it looks, the higher its score.

3. Check the top

Investigators review the highest scores first.

Fraud labels are used only to pick settings and to test the models.

PR-AUC is our main score. It goes from 0 to 1. Higher is better. A random guess scores about 0.001.

The data

284,807 card transactions over 2 days. Only 492 are fraud.

Split	Transactions	Frauds	Fraud rate
Train	170,884	360	0.21%
Validation	56,961	57	0.10%
Test	56,962	75	0.13%

We split by time

Train on the past. Test on the future. This is how a bank would really use it.

Train: model learns (normal only). Validation: pick settings. Test: final exam, used once.

What we found in the data

Not bell-shaped

Tests say none of the 28 features is normally distributed.

Long tails

Some features have extreme values far from the middle.

Clear signals

Fraud looks different on some features, such as V14.

So what?

One bell curve is too simple. We try a mix of several bell curves instead.

We cleaned the inputs

Better inputs gave a better score.

Version	What we did	PR-AUC
Raw	Used the data as given	0.043
Log amount	Shrunk huge amounts and added time of day	0.047
+ Distance	Added how far a point is from the centre	0.059
Final	Dropped 3 features with no signal	0.069

Scores are on validation data, using a simple 3-curve GMM.

Model 1: Gaussian Mixture (GMM)

What it is

A mix of bell curves that describes normal transactions.

How it scores

Low chance of being normal means a high fraud score.

How we set it up

We tried 120 settings. We kept the one that found fraud best on validation data.

Final choice

12 bell curves, with extra smoothing to keep it stable.

The setting that fits normal data best was not the best at finding fraud.

GMM results on the test set

0.41

PR-AUC. A random guess is 0.001.

69%

of frauds caught, with only 0.5% false alarms.

18%

of alerts are real fraud when we check 285 transactions.

Weak spots

Small, low-value fraud looks normal, so it is missed. Few frauds make scores jump around: 57 in validation, 75 in test. Validation (0.20) and test (0.41) scores differ a lot.

All models side by side

Model	Idea	PR-AUC	Frauds caught*
PCA	Rebuild and compare	0.74	77%
LOF	Compare with neighbours	0.45	72%
GMM	Chance of being normal	0.41	69%
One-Class SVM	Boundary around normal	0.13	69%
Isolation Forest	Isolate with random cuts	0.06	35%

*Share of the 75 test frauds caught with only 0.5% false alarms.

Winner: PCA
It rebuilds normal transactions well and fraud badly. With so few frauds, LOF and GMM are too close to call.

What this means for a bank

PCA, if we allow…	Alerts	Frauds found	Alerts that are fraud
Few false alarms (0.1%)	141	57 of 75	40%
Some false alarms (0.5%)	390	58 of 75	15%
Many false alarms (1.0%)	750	58 of 75	8%

More alerts find almost no extra fraud. They just add work for reviewers.

Limits

Only 75 test frauds. Only 2 days of data.
One dataset.

Next: Phase 2

Train an autoencoder (a neural network) and compare it with GMM and PCA.